TRANSMISSION/TRANSAXLE

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Abbreviations

AAS	Active Adaptive Shift
ABDC	After Bottom Dead Center
ABS	Antilock Brake System
ACC	Accessories
AFS	Adaptive Front lighting System
ALC	Auto Level Control
ALR	Automatic Locking Retractor
ATDC	After Top Dead Center
ATF	Automatic Transaxle Fluid
ATX	Automatic Transaxle
BBDC	Before Bottom Dead Center
BCM	Body Control Module
BDC	Bottom Dead Center
BTDC	Before Top Dead Center
CAN	Controller Area Network
CCM	Comprehensive Component Monitor
CKP	Crankshaft Position
CM	Control Module
CMDTC	Continuous Memory Diagnostic Trouble Code
CMP	Camshaft Position
CMU	Connectivity Master Unit
CPU	Central Processing Unit
DAB	Digital Audio Broadcasting
DC	Drive Cycle
DEF	Defroster
DRSS	Distance Recognition Support System
DSC	Dynamic Stability Control
EBD	Electronic Brakeforce Distribution
EEPROM	Electrically Erasable Programmable Read-Only Memory
EHPAS	Electro Hydraulic Power Assist Steering
ELR	Emergency Locking Retractor
EPS	Electric Power Steering
ESS	Emergency Stop signal System
ESU	Electrical Supply Unit
EX	Exhaust
FBCM	Front Body Control Module
FSC	Forward Sensing Camera
GPS	Global Positioning System
HBC	High Beam Control
HF/TEL	Hands-Free Telephone
HI	High
HLA	Hydraulic Lash Adjuster
HLA	Hill Launch Assist
HP	High Pressure
HS	High Speed
HU	Hydraulic Unit
IDEVA	Intake stroke EGR using Double Exhaust Valve Actuation system
IDS	Integrated Diagnostic Software
IG	Ignition
IN	Intake
INT	Intermittent
KOEO	Key On Engine Off
KOER	Key On Engine Running
L.H.D.	Left Hand Drive
LCD	Liquid Crystal Display
LDWS	Lane Departure Warning System
LED	Light Emitting Diode
LF	Left Front

LH	Left Hand
LO	Low
LP	Low Pressure
LR	Left Rear
M	Motor
MAX	Maximum
MIN	Minimum
MRCC	Mazda Radar Cruise Control
MS	Middle speed
MTX	Manual Transaxle
NVH	Noise, Vibration, Harshness
OCV	Oil Control Valve
ODDTC	On-demand Diagnostic Trouble Code
P/W CM	Power Window Control Module
PAD	Passenger Air Bag Deactivation
PCV	Positive Crankcase Ventilation
PDS	Portable Diagnostic Software
PID	Parameter Identification
POWER MOS FET	Power Metal Oxide Semiconductor Field Effect Transistor
PSD	Power Sliding Door
PTC	Positive Temperature Coefficient
R.H.D.	Right Hand Drive
RBCM	Rear Body Control Module
RCTA	Rear Crossing Traffic Alert
RDP	Run Dry Prevention
RDS	Radio Data System
REC	Recirculate
RES	Rear Entertainment System
RF	Right Front
RH	Right Hand
RR	Right Rear
SAS	Sophisticated Air Bag Sensor
SCBS	Smart City Brake Support
SCR	Secondary Collision Reduction
SST	Special Service Tool
SW	Switch
TAU	Tuner and Amp Unit
TCS	Traction Control System
TDC	Top Dead Center
TFT	Transaxle Fluid Temperature
TNS	Tail Number Side Lights
TPMS	Tire Pressure Monitoring System
USB	Universal Serial Bus
VBC	Variable Boost Control
VENT	Ventilation
W/M	Workshop Manual
WGN	Wagon
1GR	First Gear
2GR	Second Gear
2WD	2-Wheel Drive
3GR	Third Gear
4GR	Fourth Gear
4SD	4 Door Sedan
4WD	4-Wheel Drive
5GR	Fifth Gear
5HB	5 Door Hatchback
6GR	Sixth Gear

Features

Clutch, manual transaxle

- A newly developed, F65M-R-type manual transaxle (SKYACTIV-MT) with 5 forward gears and 1 reverse gear has been adopted for the SKYACTIV-G 1.5.
- A newly developed, F66M-R-type manual transaxle (SKYACTIV-MT) with 6 forward gears and 1 reverse gear has been adopted for the SKYACTIV-G 1.5 and SKYACTIV-D 1.5.

Automatic transaxle

- A newly developed, electronically controlled automatic transaxle (SKYACTIV-DRIVE) with 6 forward gears, 1 reverse gear, and a TCC mechanism has been adopted. (CW6A-EL-type for SKYACTIV-G 1.3 and SKYACTIV-G 1.5, EW6A-EL-type for SKYACTIV-D 1.5)
- An ideal automatic transaxle has been realized by pursuing fuel economy improvement, direct feel, and smooth shifting using the existing automatic transaxle as the base.
  Main features
  — Based on the adoption of a full range lock-up mechanism, a contribution to low fuel consumption and the direct feel of a manual transaxle vehicle have been realized.
  — A control valve body has been adopted with an integrated TCM, and the improved responsiveness and precision smooth shifting have been realized.

Specifications

Clutch [F65M-R]

Item			Specifications
Engine type			**SKYACTIV-G 1.5**
Clutch control			Hydraulic
Clutch pedal	Type		Suspended
	Pedal ratio		4.7
	Full stroke (reference)	(mm {in})	130.2 {5.126}
Clutch master cylinder inner diameter		(mm {in})	15.87 {0.6248}
Clutch release cylinder inner diameter		(mm {in})	20.64 {0.8126}
Clutch cover	Spring type		Diaphragm
	Set load	(N {kgf, lbf})	5,200 {530.3, 1169}
Clutch disc	Outer diameter	(mm {in})	200 {7.87}
	Inner diameter	(mm {in})	132 {5.20}
Clutch fluid type			SAE J1703 or FMVSS116 DOT-3 or DOT-4

Clutch [F66M-R]

Item			Specifications	
Engine type			**SKYACTIV-G 1.5**	**SKYACTIV-D 1.5**
Clutch control			Hydraulic	Hydraulic
Clutch pedal	Type		Suspended	Suspended
	Pedal ratio		4.7	4.7
	Full stroke (reference)	(mm {in})	130.2 {5.126}	140.6 {5.535}
Clutch master cylinder inner diameter		(mm {in})	15.87 {0.6248}	15.87 {0.6248}
Clutch release cylinder inner diameter		(mm {in})	20.64 {0.8126}	20.64 {0.8126}
Clutch cover	Spring type		Diaphragm	Diaphragm
	Set load	(N {kgf, lbf})	5,200 {530.3, 1169}	6,800 {693.4, 1529}
Clutch disc	Outer diameter	(mm {in})	200 {7.87}	228 {8.98}
	Inner diameter	(mm {in})	132 {5.20}	155 {6.10}
Clutch fluid type			SAE J1703 or FMVSS116 DOT-3 or DOT-4	SAE J1703 or FMVSS116 DOT-4

Manual Transaxle [F65M-R]

Item			Specifications
Engine type			**SKYACTIV-G 1.5**
Transaxle control			Floor-shift
Operation system			Cable
Shift assist	Forward		Synchromesh
	Reverse		Constant mesh
Gear ratio	1GR		3.583
	2GR		1.904
	3GR		1.218
	4GR		0.918
	5GR		0.717
	Reverse		3.454
Final gear ratio			3.619
Manual transaxle oil	Grade		API service GL-4
	Viscosity	All season	SAE 75W-80
	Capacity (approx. quantity)	(L {US qt, Imp qt})	1.59 {1.68, 1.40}

Manual Transaxle [F66M-R]

Item			Specifications	
Engine type			**SKYACTIV-G 1.5**	**SKYACTIV-D 1.5**
Transaxle control			Floor-shift	
Operation system			Cable	
Shift assist	Forward		Synchromesh	
	Reverse		Constant mesh	
Gear ratio	1GR		3.583	3.230
	2GR		1.904	1.652
	3GR		1.290	1.088
	4GR		0.972	0.775
	5GR		0.795	0.580
	6GR		0.645	0.490
	Reverse		3.454	3.454
Final gear ratio			With i-stop : 4.388 Without i-stop : 4.105	3.850
Manual transaxle oil	Grade		API service GL-4	API service GL-4
	Viscosity	All season	SAE 75W-80	SAE 75W-80
	Capacity (approx. quantity)	(L {US qt, Imp qt})	1.59 {1.68, 1.40}	1.64 {1.73, 1.44}

Automatic Transaxle [CW6A-EL]

Item			Specifications	
Engine type			**SKYACTIV-G 1.3**	**SKYACTIV-G 1.5**
Gear ratio		1GR	3.529	
		2GR	2.025	
		3GR	1.348	
		4GR	1.000	
		5GR	0.742	
		6GR	0.594	
		Reverse	2.994	
Final gear ratio			4.060	4.319
ATF	Type		ATF FZ	
	Capacity (Approx. quantity)	(L {US qt, Imp qt})	7.2 {7.6, 6.3}	
Torque converter stall torque ratio			2.22	

Item		Specifications	
Engine type		**SKYACTIV-G 1.3**	**SKYACTIV-G 1.5**
Hydraulic system (Number of drive/ driven gear plates)	Low clutch	3/3	
	High clutch	4/4	
	Low and reverse brake	4/4	
	2-6 brake	3/3	
	R-3-5 brake	2/2	
Front planetary gear (Number of teeth)	Front sun gear	34	
	Front pinion gear	26	
	Front internal gear	86	
Rear planetary gear (Number of teeth)	Rear sun gear	37	
	Rear pinion gear	26	
	Rear internal gear	89	
Reduction planetary gear (Number of teeth)	Reduction sun gear	49	
	Reduction pinion gear No.1	22	
	Reduction pinion gear No.2	23	
	Reduction internal gear	100	
Primary gear (number of teeth)		41	
Secondary gear (number of teeth)		46	
Output gear (number of teeth)		21	20
Ring gear (number of teeth)		76	77

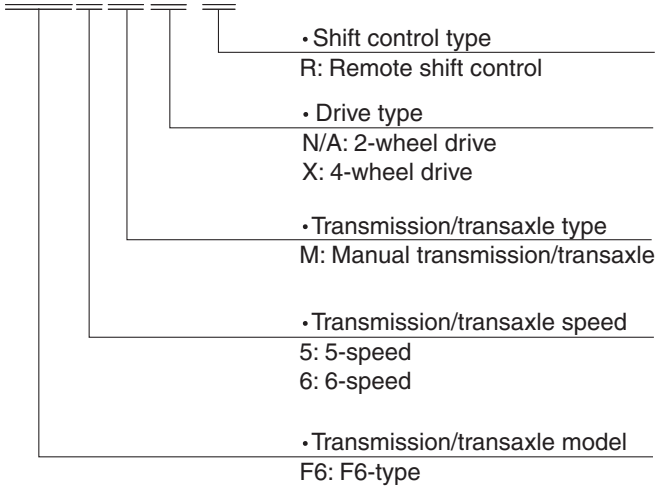
Automatic Transaxle [EW6A-EL]

Item			Specifications
Engine type			**SKYACTIV-D 1.5**
Gear ratio		1GR	3.552
		2GR	2.022
		3GR	1.452
		4GR	1.000
		5GR	0.708
		6GR	0.599
		Reverse	3.893
Final gear ratio			3.389
ATF	Type		ATF FZ
	Capacity (Approx. quantity)	(L {US qt, Imp qt})	7.8 {8.2, 6.9}
Torque converter stall torque ratio			1.82
Hydraulic system (Number of drive/ driven gear plates)	Low clutch		5/5
	High clutch		5/5
	Low and reverse brake		3/5
	2-6 brake		3/5
	R-3-5 brake		2/4
Front planetary gear (Number of teeth)	Front sun gear		38
	Front pinion gear		29
	Front internal gear		97
Rear planetary gear (Number of teeth)	Rear sun gear		40
	Rear pinion gear		27
	Rear internal gear		95
Reduction planetary gear (Number of teeth)	Reduction sun gear		68
	Reduction pinion gear		21
	Reduction internal gear		109
Primary gear (number of teeth)			56
Secondary gear (number of teeth)			59
Output gear (number of teeth)			23
Ring gear (number of teeth)			74

Identification

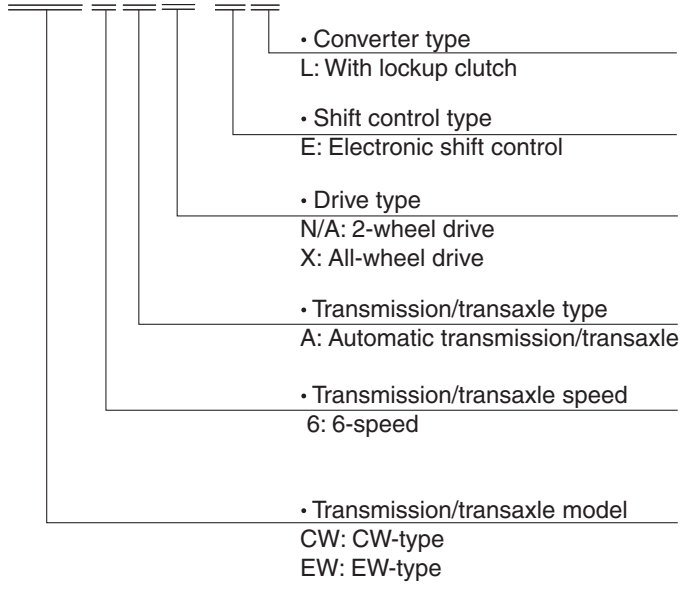
Manual Transaxle

F66MX-R



Automatic Transaxle

CW6AX-EL



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